Unit 4 Topic Questions

1. An object of mass $m = 0.5\ \text{kg}$ is moving in the xy-plane with velocity components of $v_x = 3.0\ \text{m/s}$ and $v_y = -4.0\ \text{m/s}$. What is the magnitude of the object's momentum?

(A) $0.5\ \text{kg·m/s}$

(B) $2.5\ \text{kg·m/s}$ ✓

(C) $3.5\ \text{kg·m/s}$

(D) $5.0\ \text{kg·m/s}$

Answer B

Correct. The magnitude of the object's momentum is determined by the product of its mass and the magnitude of its velocity.

$$\left|\overrightarrow{p}\right| = m\left|\overrightarrow{v}\right|$$

$$\left|\overrightarrow{v}\right| = \sqrt{v_x^2 + v_y^2}$$

$$\left|\overrightarrow{v}\right| = \sqrt{(3.0\ \text{m/s})^2 + (-4.0\ \text{m/s})^2} = 5.0\ \text{m/s}$$

$$\left|\overrightarrow{p}\right| = m\left|\overrightarrow{v}\right| = (0.5\ \text{kg})(5.0\ \text{m/s}) = 2.5\ \text{kg·m/s}$$

2. A moving block of mass m_0 has kinetic energy K_0 and momentum of magnitude p_0. A second block of mass $2m_0$ is moving with the same kinetic energy K_0. What is the magnitude of the momentum of the second block?

(A) $2p_0$

(B) $\frac{2p_0}{\sqrt{2}}$ ✓

(C) p_0

(D) $\frac{p_0}{\sqrt{2}}$

Answer B

Correct. The second block has a velocity which is a factor of $\frac{1}{\sqrt{2}}$ of the velocity of the first block in order to have the same kinetic energy. The momentum of the second block therefore will be a factor of $\frac{2}{\sqrt{2}}$ of the momentum of the first block due to the differences in mass and velocity compared to the first block.

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3. The kinetic energy of a moving object changes from K_0 to $4K_0$. By what factor does the magnitude of the object's momentum change?

(A) 1

(B) 2 ✓

(C) 4

(D) 16

Answer B

Correct. The momentum of the object is proportional to the velocity of the object. For a constant mass, the velocity of the object is proportional the square root of its kinetic energy and therefore doubles.

4. A block with a mass of $2.0\ \text{kg}$ starts from rest and begins to move along the x-axis at time $t = 0$ according to the equation $x = \frac{1}{2}t^3$, where x is the block's position, in meters, and t is the time, in seconds. What is the momentum of the block at time $t = 4.0\ \text{s}$?

(A) $16\ \text{kg} \cdot \text{m/s}$

(B) $24\ \text{kg} \cdot \text{m/s}$

(C) $48\ \text{kg} \cdot \text{m/s}$ ✓

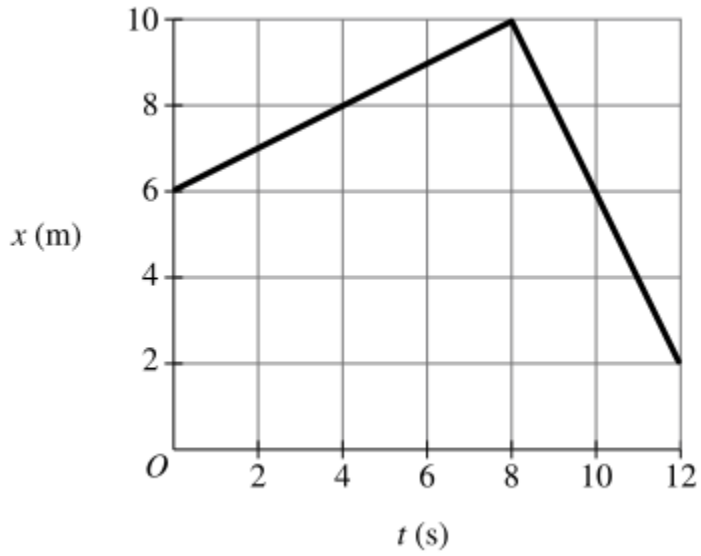
(D) $64\ \text{kg} \cdot \text{m/s}$

Answer C

Correct. The velocity of the block is determined using the derivative of the position function as $v = \frac{3}{2}t^2$ and evaluating at $t = 4.0\ \text{s}$ yields $v = 24\ \text{m/s}$. The momentum is determined as the product of the mass and the velocity resulting in $p = 48\ \text{kg}\cdot\text{m/s}$

Unit 4 Topic Questions

5.



The graph shows the position x as a function of time t for a cart of mass $m = 3.0\ \text{kg}$ that moves along the x-axis. What is the magnitude of the change in the cart's momentum between $t = 4\ \text{s}$ and $t = 10\ \text{s}$?

(A) $1.0\ \text{kg·m/s}$

(B) $2.5\ \text{kg·m/s}$

(C) $4.5\ \text{kg·m/s}$

(D) $7.5\ \text{kg·m/s}$ ✓

Answer D

Correct. The velocity of the object can be calculated from the slope of the graph at the two given times. The magnitude of the change in momentum is calculated as: $|\Delta p| = |m(\Delta v)|$, $|\Delta p| = |3\ \text{kg}(-2\ \text{m/s} - 0.5\ \text{m/s})|$, $|\Delta p| = 7.5\ \text{kg} \cdot \text{m/s}$.

6. A block slides along a surface with a speed v_0 and a momentum of magnitude p_0. It collides with and sticks to an identical block that is initially at rest. Immediately after the collision, the two blocks slide together with a speed of $\frac{1}{2}v_0$. What is the magnitude of the momentum of the two-block system immediately after the collision?

(A) $\frac{1}{4}p_0$

(B) $\frac{1}{2}p_0$

(C) p_0 ✓

(D) $2p_0$

Unit 4 Topic Questions

Answer C

Correct. The momentum of the system is proportional to the product of the mass and the velocity of the system. Because the mass doubles and the velocity is halved, the momentum of the system is the same as the momentum of the single block before the collision.

7. Two moving objects each have momenta of equal magnitude p_0. Object 1 is more massive than Object 2. Which of the following claims about the speed of Object 2 is true?

(A) It is greater than the speed of Object 1. ✓

(B) It is equal to the speed of Object 1.

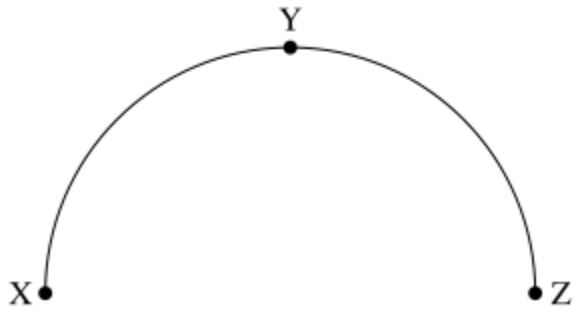
(C) It is less than the speed of Object 1.

(D) It could be greater than, equal to, or less than the speed of Object 1 depending on the directions of the momenta.

Answer A

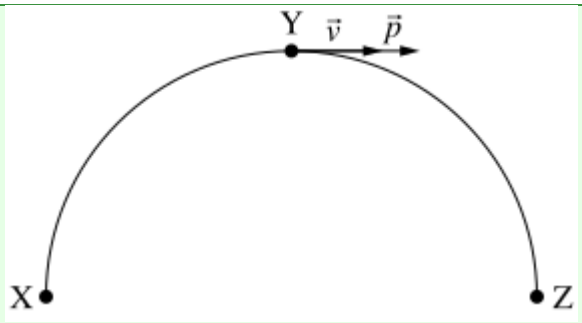
Correct. The momentum is defined by $\vec{p} = m\vec{v}$. The magnitudes of the momenta are then $p_0 = m_1 v_1$ and $p_0 = m_2 v_2$. Because $m_1 > m_2$, it follows that $v_1 < v_2$.

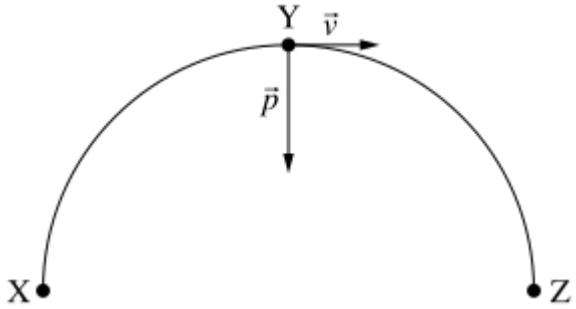
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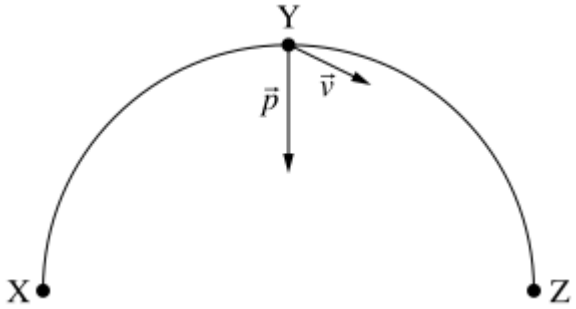


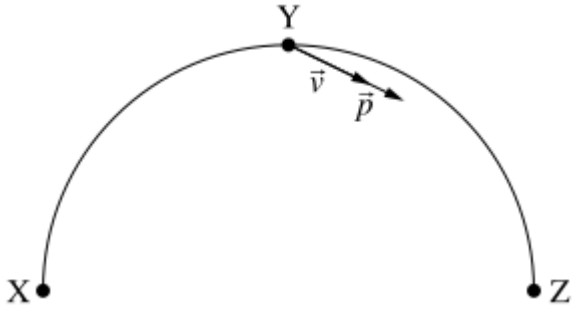
A semicircular path connects three points labeled X, Y, and Z, as shown in the figure. A particle is initially at point X and moves along the path with a constant speed toward point Z. When the particle is at point Y, as shown in the figure, which of the following diagrams correctly represents the directions of the velocity $\vec{v}$ and momentum $\vec{p}$ of the object?

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(A) 

(B) 

(C) 

(D) 

Answer A

Correct. The velocity $\overrightarrow{v}$ must be tangent to the circular path and the momentum $\overrightarrow{p}$ is parallel to the velocity.

Unit 4 Topic Questions

9. A student is pulling a cart full of water horizontally on flat ground at a constant velocity. The cart has a small hole through which the water slowly leaks out of the cart. For the system consisting of the cart and the water within the cart, how is the magnitude of the momentum of the system changing, if at all?

(A) The magnitude of the momentum is increasing.

(B) The magnitude of the momentum remains the same.

(C) The magnitude of the momentum is decreasing. ✓

(D) The answer cannot be determined without knowing the cart's mass and the mass of water that leaks through the hole each second.

Answer C

Correct. Momentum is defined as $\overrightarrow{p} = m\overrightarrow{v}$. In this situation the velocity is constant, and mass is leaving the system, therefore the magnitude of the momentum is decreasing.

10. A 50-kg student sitting in a rolling chair at rest pushes against a wall, which applies a $10\ \text{N}\cdot\text{s}$ horizontal impulse to the student. Later, a 40-kg student is at rest in the same rolling chair and catches a 10-kg ball while applying a $10\ \text{N}\cdot\text{s}$ horizontal impulse to the ball. Which of the following statements best describes the students' final speeds and provides valid reasoning supporting the claim?

(A) Both students have the same final speed because the $50\ \text{kg}$ student-chair system and the $10\ \text{kg}$ ball-$40\ \text{kg}$ student-chair system have the same mass and experience impulses of the same magnitude.

(B) Both students have the same final speed because each ball-chair-student system experiences the same impulse regardless of the mass of the system.

(C) The 40-kg student has a greater final speed because the 40-kg student-chair system has less mass than the 50-kg student-chair system. ✓

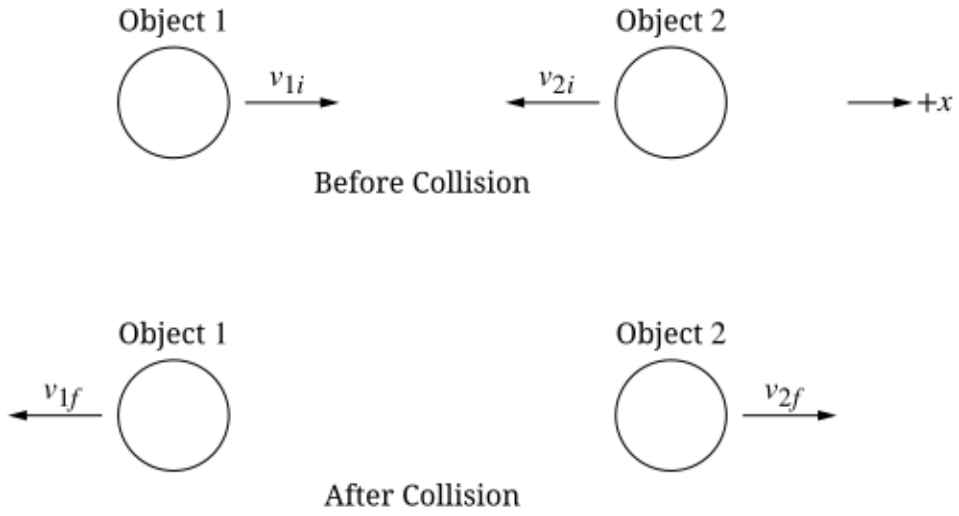
(D) The 40-kg student has a greater final speed because the ball transfers all of its momentum to the student when it is caught.

Answer C

Correct. In the interaction between the 40-kg student and the ball, both the ball and the 40-kg student-chair system experience $10\ \text{N}\cdot\text{s}$ of impulse. This means that the 40-kg student-chair system has the same impulse acting on it as the 50-kg student-chair system. When two systems of differing mass experience the same impulse, the less massive system will have the greater change in velocity. Consequently the 40-kg student-chair system, and therefore the 40-kg student, will have a greater final speed

Unit 4 Topic Questions

11.



Object 1 of mass m_1 and Object 2 of mass m_2 move along the x-axis toward each other with speeds v_{1i} and v_{2i}, respectively. The objects collide and then move away from each other with speeds v_{1f} and v_{2f}, respectively, as shown in the figure. There are no external forces acting on the two-object system. What is the magnitude of the impulse exerted on Object 1 by Object 2 during the collision?

(A) $m_1|v_{1f} - v_{1i}|$

(B) $m_1(v_{1f} + v_{1i})$ ✓

(C) $m_1|v_{2f} - v_{2i}|$

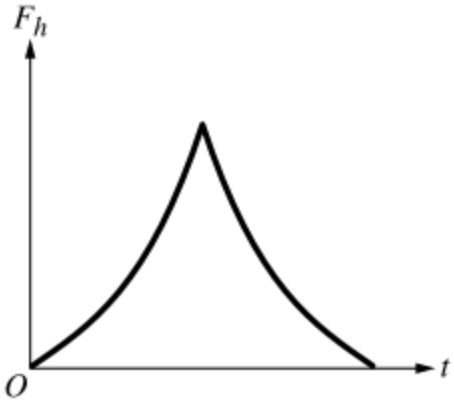
(D) $m_1(v_{2f} + v_{2i})$

Answer B

Correct. The magnitude of the impulse on Object 1 is equal to the magnitude of the change in momentum of Object 1. The change in momentum is defined as $\Delta p = m\Delta v$, where the change in velocity has a magnitude $|\Delta v| = |v_{1f} - (-v_{1i})| = v_{1f} + v_{1i}$.

Unit 4 Topic Questions

12.



A piece of clay is thrown against a hard wall. The graph shows the horizontal component F_h of the net force exerted on the clay during the collision as a function of time t. How would the maximum of F_h, the duration of the collision, and the area under the graph change if an identical piece of clay was instead thrown against a padded wall with the same velocity? Assume that both collisions are perfectly inelastic.

	Maximum of F_h	**Duration of collision**	**Area under graph**
A	Increases	Decreases	Increases
B	Decreases	Increases	Remains the same
C	Increases	Decreases	Remains the same
D	Decreases	Remains the same	Decreases

(A) A

(B) B ✓

(C) C

(D) D

Answer B

Correct. The clay impacts each surface with the same initial momentum and is brought to rest. Therefore, the impulse experienced by the clay in each case is equal and the area under the graph remains the same. In one dimension:

Unit 4 Topic Questions

$$\Delta p_1 = \int_{t_i}^{t_{f_1}} F_1 dt$$

$$\Delta p_2 = \int_{t_i}^{t_{f_2}} F_2 dt$$

$$\Delta p_1 = \Delta p_2 \rightarrow \int_{t_i}^{t_{f_1}} F_1 dt = \int_{t_i}^{t_{f_2}} F_2 dt$$

The collision with the padded surface will have a greater duration and thus the maximum force experienced by the clay will decrease in order for the area under the graph to remain the same.

13. A force F that varies with time t is exerted on an object. The force is described by the equation $F(t) = F_0 e^{-bt}$ where F_0 and b are positive constants. If the object is at rest at time $t = 0$, the momentum of the object a very long time later is most nearly

(A) zero

(B) bF_0

(C) $\frac{F_0}{b}$ ✓

(D) $-\frac{F_0}{b}$

Answer C

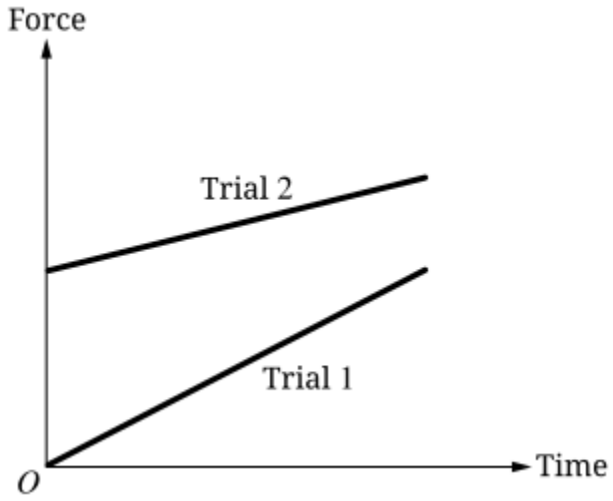
Correct. The change in momentum of the object is equal to the impulse delivered by the force.

$$\Delta p = \int_0^\infty F dt = \int_0^\infty F_0 e^{-bt} dt = -\frac{F_0}{b} e^{-bt}\Big|_0^\infty$$

$$= 0 - \left(-\frac{F_0}{b}\right) = \frac{F_0}{b}$$

Unit 4 Topic Questions

14.



Students are conducting an experiment on a block that is free to move along a straight, horizontal track with negligible friction. They conduct two trials, Trial 1 and Trial 2, in which the block is initially at rest and a time-varying horizontal force is applied to the block in a direction parallel to the track. The applied force as a function of time is shown in the graph for each trial, both of which last the same amount of time. During Trial 1, the students do work W_1 on the block and apply an impulse J_1. During Trial 2, they do work W_2 on the block and apply an impulse J_2. How do these values of work and impulse compare to each other?

	Work	Impulse
A	$W_1 < W_2$	$J_1 < J_2$
B	$W_1 < W_2$	$J_1 > J_2$
C	$W_1 > W_2$	$J_1 < J_2$
D	$W_1 > W_2$	$J_1 > J_2$

(A) A ✓

(B) B

(C) C

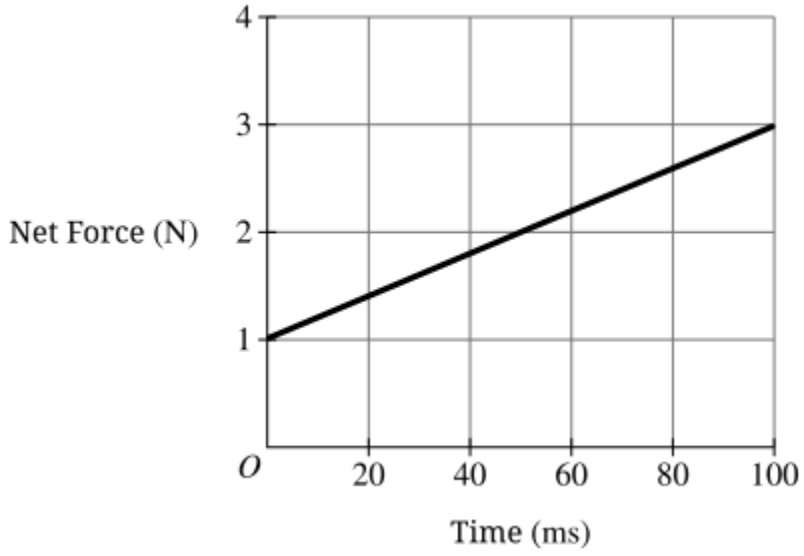
(D) D

Answer A

Correct. The impulse is found from the area under the graph, which is greater in Trial 2. Because the force is greater at any point of time in Trial 2, there is a greater acceleration at every time. Therefore, the block in Trial 2 will reach a greater final speed and thus a greater kinetic energy, meaning that more work was done on it.

Unit 4 Topic Questions

15.



A lab cart is traveling along a straight, horizontal track with an unknown initial velocity. The graph shows the net force on the cart as a function of time during a short time interval. Does the graph provide sufficient information to determine the change in momentum of the cart during the time interval shown, and what evidence supports the claim?

(A) Yes, because the change in momentum is equal to the area under the curve of the force-versus-time graph. ✓

(B) Yes, because the change in momentum is equal to the slope of the force-versus-time graph.

(C) No, because the initial momentum is unknown and therefore the change in momentum cannot be determined.

(D) No, because the force is not constant and therefore the change in momentum cannot be determined.

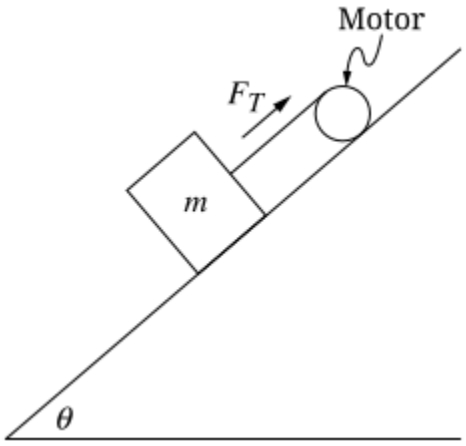
Answer A

Correct. The change in momentum is equal to the impulse, which is defined as

$$\vec{J} = \int_{t_1}^{t_2} \vec{F}_{\text{net}}(t)\,dt = \Delta\vec{p}.$$

Unit 4 Topic Questions

16.



A block of mass m can move with negligible friction on a ramp that is at an angle θ from the horizontal, as shown in the figure. Starting at time $t = 0$, the motor creates a varying string tension F_T given by $F_T = 2kt$, where k is a positive constant. If the positive direction is taken to be up the ramp, what is the magnitude of the impulse exerted on the block between time $t_1 > 0$ and a later time t_2 ?

(A) $k\left({t_2}^2 - {t_1}^2\right)$

(B) $k{t_2}^2 - mg\sin\theta(t_2)$

(C) $k\left({t_2}^2 - {t_1}^2\right) - mg\sin\theta(t_2 - t_1)$ ✓

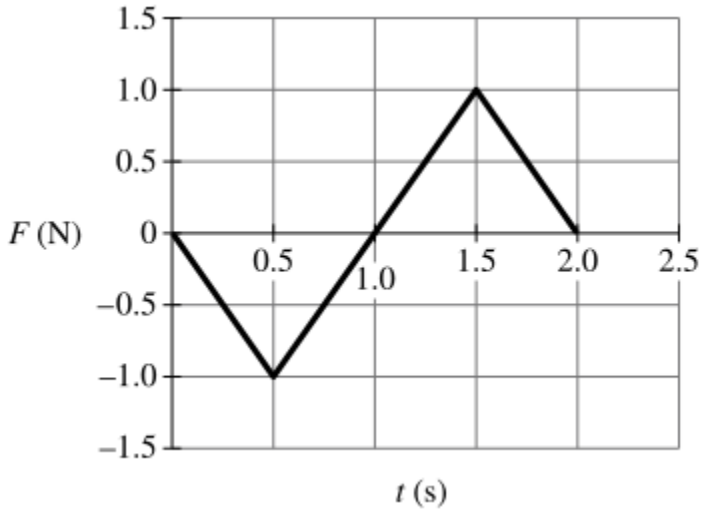
(D) $k\left({t_2}^2 - {t_1}^2\right) - mg(t_2 - t_1)$

Answer C

Correct. Impulse is defined as $\vec{J} = \int_{t_1}^{t_2} \vec{F}_{\text{net}}\,dt$. The net force exerted on the block is the vector sum of the tension force and the component of the weight down the incline. Therefore $\vec{J} = \int_{t_1}^{t_2} (2kt - mg\sin\theta t)dt$ which yields $\vec{J} = k\left({t_2}^2 - {t_1}^2\right) - mg\sin\theta(t_2 - t_1)$.

Unit 4 Topic Questions

17.



The graph shows the net force F exerted on an object as a function of time t. How does the impulse J_1 exerted on the object from $t = 0$ s to $t = 1$ s compare to the impulse J_2 exerted on the object from $t = 1$ s to $t = 2$ s ?

(A) $J_1 = -J_2$ ✓

(B) $J_1 = J_2$

(C) $|J_1| < |J_2|$

(D) $|J_1| > |J_2|$

Answer A

Correct. Impulse is the area under the curve and, in this case, the part of the curve that is negative is exactly the negative of the part of the curve that is positive.

18. A toy water launcher, initially full of water, can slide with negligible friction on a horizontal surface. Water is ejected horizontally from the launcher with a constant speed relative to the launcher. The mass of water ejected from the launcher over any time interval of length Δt is constant as well. Which of the following statements correctly describes how the launcher moves, if at all, and provides a correct justification?

(A) The launcher will remain in place because there is no horizontal force acting on the launcher.

(B) The launcher will move at a constant speed because the water is ejected at a constant speed.

(C) The launcher will move at a constant acceleration because the system of the launcher and remaining water experiences the same impulse during any time interval of length Δt.

(D) The launcher will move at an increasing acceleration because the mass of the system of the launcher and remaining water is decreasing, and the system experiences the same impulse during any time interval of length Δt. ✓

Unit 4 Topic Questions

Answer D

Correct. By Newton's third law, because the water is ejected by the launcher, the water also pushes back on the launcher. The ejection of water also decreases the total mass of the launcher and remaining water, so by Newton's second law, the acceleration of the launcher will increase.

19. In three different scenarios, a student of mass $3m$ pulls on a lightweight rope attached to a block of mass m. The student and the block are both initially at rest in each scenario. In Scenario 1, the student stands on the shore of a frozen lake and the block is on the ice. After the student starts pulling on the rope in Scenario 1, the block slides with negligible friction while the student remains in the same location on the shore. In Scenario 2, both the student and the block are on the ice. After the student starts pulling on the rope in Scenario 2, both the student and the block slide with negligible friction. In Scenario 3, the block is at rest on the shore while the student is on the ice. After the student starts pulling on the rope in Scenario 3, the student slides with negligible friction while the block remains in the same location on the shore. The final velocity of the block relative to the student is the same in each scenario. For which scenario, if any, is the final speed of the center of mass of the block-student system the greatest?

(A) Scenario 1

(B) Scenario 2

(C) Scenario 3 ✓

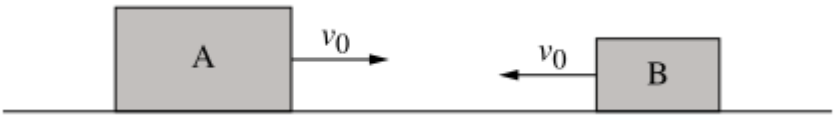
(D) The speed of the center of mass is the same in all three scenarios.

Answer C

Correct. The center of mass velocity is zero for Scenario 2. More mass is moving in Scenario 3, when the student moves across the ice, than in Scenario 1. Therefore the center of mass velocity for Scenario 3 is the greatest.

Unit 4 Topic Questions

20.



Trial	Center of mass speed after collision	Impulse exerted on Block B
1	v_1	J_1
2	v_2	J_2

Blocks A and B can move along a horizontal surface with negligible friction, as shown in the figure. The mass of Block A is three times the mass of Block B. In two separate experimental trials, the blocks are initially moving toward each other, each with a speed of v_0. In Trial 1, Block A comes to rest after the collision, while in Trial 2 the blocks stick together. The table shows the speed of the center of mass of the two-block system after the collision in each trial, as well as the magnitude of the impulse exerted on Block B during each collision. Which of the following correctly compares these values?

	Center of mass speeds	**Magnitude of the impulses exerted on Block B**
A	$v_1 = v_2$	$J_1 > J_2$
B	$v_1 = v_2$	$J_1 = J_2$
C	$v_1 > v_2$	$J_1 > J_2$
D	$v_1 > v_2$	$J_1 = J_2$

(A) A ✓

(B) B

(C) C

(D) D

Answer A

Correct. Since there is no net external force on the two-block system in either trial, the velocity of the center of mass of the system is unchanged during either collision. In Trial 1, Block A experiences a greater momentum change than in Trial 2, and therefore Block B does as well since the total momentum change is zero for the two-block system. Since impulse equals change in momentum, Block B experiences a greater impulse in Trial 1 than in Trial 2.

Unit 4 Topic Questions

21.



Two blocks A and B have masses m and $2m$, respectively, and are free to move on a horizontal surface with negligible friction, as shown in the figure. Block A is initially moving with speed v_0 toward Block B, which is at rest and has a spring attached to one side. Block A then makes contact with the spring, causing the spring to compress. After the spring reaches its maximum compression, the blocks move apart and Block A loses contact with the spring. What is the speed of the center of mass of the blocks-spring system after the blocks have separated?

(A) v_0

(B) $\frac{2v_0}{3}$

(C) $\frac{v_0}{2}$

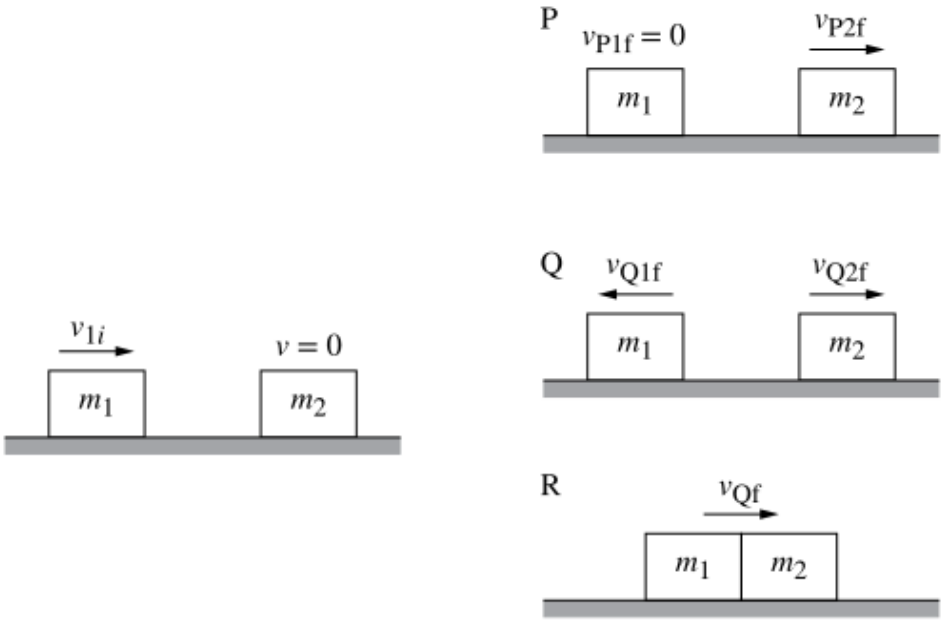
(D) $\frac{v_0}{3}$ ✓

Answer D

Correct. Because the net external force on the blocks-spring system is zero, the velocity of the center of mass of the system is the same before and after the collision. The center of mass speed is then $v_{\mathrm{cm}} = \frac{(\sum m_i v_i)}{(\sum m_i)} = \frac{mv_0}{3m} = \frac{v_0}{3}$.

Unit 4 Topic Questions

22.



A block of mass m_1 moves with initial speed v_{1i} on a horizontal surface toward a stationary block of mass m_2, as shown in the figure labeled "Before Collision." Friction between the blocks and the horizontal surface is negligible. Three possible outcomes labeled P, Q, and R, are also shown. Rank the two-block system's final center of mass speeds v_{Pcm}, v_{Qcm}, and v_{Rcm} after the three collision outcomes.

(A) $v_{\text{Pcm}} = v_{\text{Qcm}} = v_{\text{Rcm}}$ ✓

(B) $v_{\text{Qcm}} < v_{\text{Rcm}} < v_{\text{Pcm}}$

(C) $v_{\text{Rcm}} < v_{\text{Pcm}} < v_{\text{Qcm}}$

(D) $v_{\text{Pcm}} < v_{\text{Qcm}} < v_{\text{Rcm}}$

Answer A

Correct. The velocity of the center of mass will be constant in the absence of a net force acting on the two-block system. In each of these collisions, the net force on the system is zero, so the center of mass velocity after each collision is the same as the initial center of mass velocity.

Unit 4 Topic Questions

23.

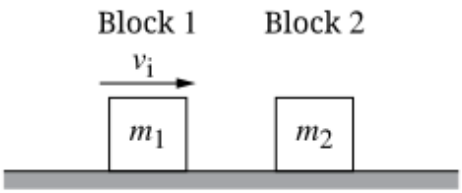


Figure 1

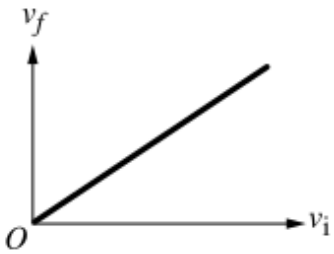


Figure 2

A collision experiment is conducted using two blocks, Block 1 and Block 2, that can slide with negligible friction along a horizontal surface. Block 1 of mass m_1 is initially moving with speed v_i toward Block 2, which has a mass m_2 and is initially at rest, as shown in Figure 1. The blocks collide, and after the collision they stick together and move with speed v_f. In multiple trials of this experiment, Block 1 is given different initial velocities. Data for v_f as a function of v_i are recorded, and the data is used to make a best-fit curve, which is a straight line as shown in Figure 2. If the experiment were repeated with Block 1 replaced by a block of mass $2m_1$, how would the best-fit line to the data change, if at all?

(A) The slope of the line would be less.

(B) The slope of the line would be greater. ✓

(C) The line would become a curve with a slope that is not constant.

(D) The line would not change.

Answer B

Correct. Conservation of momentum gives that $m_1 v_i = (m_1 + m_2) v_f$ and hence $v_f = \frac{m_1}{m_1 + m_2} v_i$. Doubling m_1 would increase the slope of the graph, because the mass ratio term in the equation approaches 1 from below as m_1 increases.

24. A 4-kg object moving to the left collides with and sticks to a 3-kg object moving to the right. Just before the collision, the lighter object is moving faster than the heavier object. Which of the following is true of the combined objects immediately after the collision?

(A) They must be moving to the left.

(B) They must be moving to the right.

(C) They must be at rest.

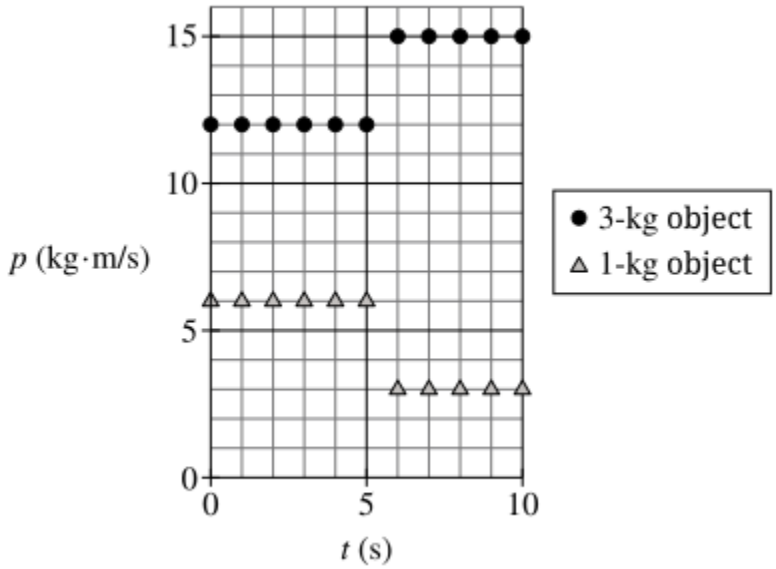
(D) The combined objects' direction of motion, if any, cannot be determined without knowing the ratio of the speeds of the objects before the collision. ✓

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Answer D

Correct. Without knowing the ration of the speeds of the objects, the direction of the total momentum of the two-object system cannot be determined. Hence it is impossible to determine in which direction the combined objects move.

25.



A 1-kg object moving at $6\ \text{m/s}$ collides with a 3-kg object moving at $4\ \text{m/s}$ in the same direction. Data for the momentum p as a function of time t for both objects is shown in the graph. Which of the following statements best describes this collision?

(A) The momentum of the two-object system is constant, and the magnitude of the relative velocity between the objects is the same before and after the collision. ✓

(B) The momentum of the two-object system is constant, and the magnitude of the relative velocity between the objects is different before and after the collision.

(C) The momentum of the 1-kg object is constant, and the magnitude of the relative velocity between the objects is the same before and after the collision.

(D) The momentum of the 1-kg object is not constant, and the magnitude of the relative velocity between the objects is different before and after the collision.

Answer A

Correct. The momentum of the system calculated before and after the collision is the same.
$p_i = \sum_n m_n v_n = 6\ \text{kg} \cdot \text{m/s} + 12\ \text{kg} \cdot \text{m/s} = 18\ \text{kg} \cdot \text{m/s};$

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$p_f = \sum_n m_n v_n = 3\,\text{kg} \cdot \text{m/s} + 15\,\text{kg} \cdot \text{m/s} = 18\,\text{kg} \cdot \text{m/s}$. The relative speed between the objects before they collide is $6\,\text{m/s} - 4\,\text{m/s} = 2\,\text{m/s}$. The objects velocities after the collision are $v_1 = \left(\frac{3}{1}\right)\,\text{m/s} = 3\,\text{m/s}$ and $v_3 = \left(\frac{15}{3}\right)\,\text{m/s}$, so the relative speed is still $2\,\text{m/s}$.

26. Two objects, both of mass m, are free to slide with negligible friction on a horizontal surface that lies in the plane of an xy-coordinate system. Initially one of the objects moves with a velocity v_0 in the $+x$-direction and toward the second object, which is at rest. The two objects interact for a brief time with no net external force acting on the two-object system. After the interaction the first object moves with a velocity that has positive x and y components, a magnitude v_0, and is directed at an angle θ from the $+x$-direction. What are the final x-component v_x and y-component v_y of the velocity of the object that was initially at rest?

	v_x	v_y
A	$v_0(1 - \cos\theta)$	$-v_0 \sin\theta$
B	$-v_0 \cos\theta$	$-v_0 \sin\theta$
C	v_0	$-v_0$
D	0	0

(A) A ✓

(B) B

(C) C

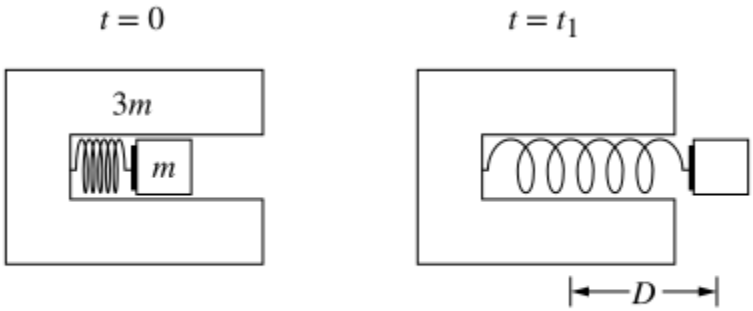
(D) D

Answer A

Correct. With no net external force exerted on the two-object system, a conserved quantity such as total momentum stays the same. The system's initial momentum is mv_0 in the $+x$-direction. The first object's final momentum has components $p_{x1} = mv_0 \cos\theta$ and $p_{y1} = mv_0 \sin\theta$. In order for the system to have the same momentum before and after the interaction, the second object's final momentum must have an x-component of $mv_0 - p_{x1} = mv_0(1 - \cos\theta)$ and a y-component of $0 - p_{y1} = mv_0 \sin\theta$. Dividing these momentum components by m results in the correct velocity components.

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27.



Note: Figure not drawn to scale.

A small block of mass m is held in place inside a larger, U-shaped block of mass $3m$, as shown in the figure. Initially, the centers of mass of the two blocks are located at the same position, and a spring attached to the larger block is compressed a distance D from its relaxed length by the smaller block. At time $t = 0$, the two blocks are released from rest. At time $t_1 > 0$, the spring is at its relaxed length, the centers of mass of the two blocks are a distance D apart, and the small block is moving to the right with a velocity v just as it loses contact with the spring. If there are no external forces exerted on the blocks-spring system, what is the distance of separation of the two centers of mass at a later time $t > t_1$?

(A) $4v(t - t_1)$

(B) $D + 4v(t - t_1)$

(C) $D + \frac{2}{3}v(t - t_1)$

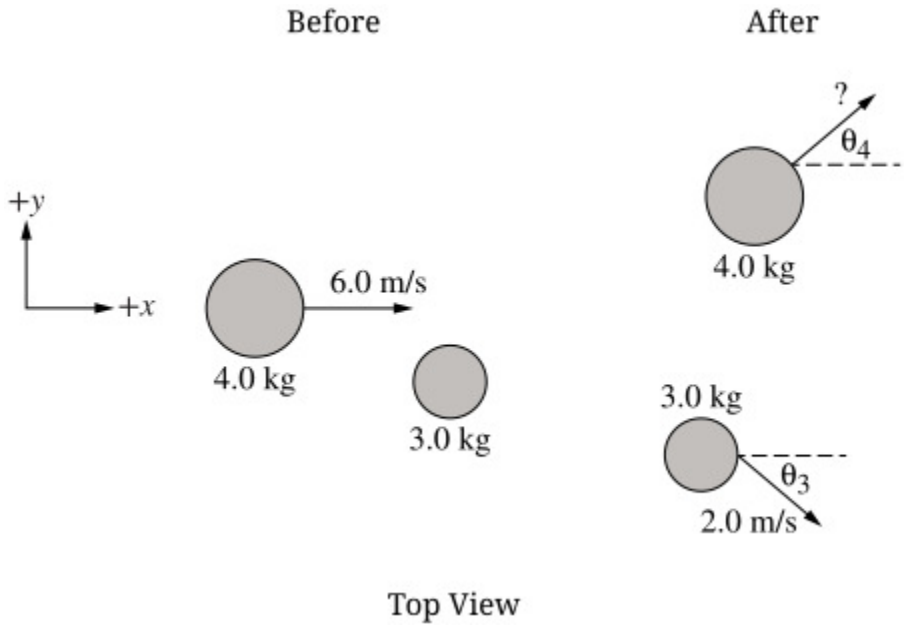
(D) $D + \frac{4}{3}v(t - t_1)$ ✓

Answer D

Correct. Conservation of momentum gives that $0 = mv + 3mv_3$, where the velocity of the $3m$ block is v_3, and therefore $v_3 = -\frac{v}{3}$. At time t_1, the blocks' centers of mass have separated a distance D. After time t_1, the additional distance the small and large blocks travel are given by $v(t - t_1)$ and $\frac{v}{3}(t - t_1)$, respectively. The total separation of the blocks' centers of mass is therefore $D + \frac{4}{3}v(t - t_1)$.

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28.



Two pucks are free to slide on a horizontal surface with negligible friction. The figure shows a top view of the pucks, one of mass $4.0\ \text{kg}$ and one of mass $3.0\ \text{kg}$, before and after they undergo an elastic collision with each other. Before the collision, the 4.0-kg puck slides at $6.0\ \text{m/s}$ in the $+x$-direction and the 3.0-kg puck is at rest. After the collision, the 3.0-kg puck moves with a speed of $2.0\ \text{m/s}$ at an unknown angle θ_3 measured clockwise from the $+x$-direction, as indicated, while the 4.0-kg puck moves at an unknown speed and at an unknown angle θ_4 measured counterclockwise from the $+x$-direction, as indicated. What is the speed of the 4.0-kg puck after the elastic collision?

(A) $1.5\ \text{m/s}$

(B) $4.0\ \text{m/s}$

(C) $4.5\ \text{m/s}$

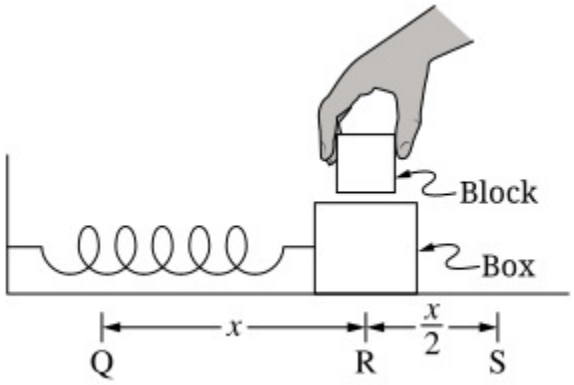
(D) $5.7\ \text{m/s}$ ✓

Answer D

Correct. Because this is an elastic collision, there is enough information to solve for the final velocity by equating the initial and final total kinetic energies of the two-puck system. It is not necessary to use conservation of momentum: $\frac{1}{2}(4\ \text{kg})(6\ \text{m/s})^2 = \frac{1}{2}(3\ \text{kg})(2\ \text{m/s})^2 + \frac{1}{2}(4\ \text{kg})(v)^2$. Solving this equation for v results in the answer.

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29.



A spring has its left end fixed to a wall and its right end attached to a box, which can move back and forth on a horizontal surface with negligible friction, as shown in the figure. When the spring is at its relaxed length the box is at position R. The spring is compressed a distance x from its relaxed length and the box is then released from rest at position Q. The box moves from position Q to position R, continues moving on to position S a distance $\frac{x}{2}$ to the right of R, and continues moving beyond position S toward the right. As the box moves, a block can be dropped onto the box from a small height just above the box. When this happens, the block sticks to the box in a perfectly inelastic collision and the two objects move together horizontally after colliding. At which labeled position(s) can the block be dropped onto the box so that the block-box system's total momentum and total kinetic energy are each the same immediately before and immediately after the collision? Neglect the block's vertical component of motion, and treat the collision as occurring instantaneously.

(A) At position Q only ✓

(B) At position R only

(C) At position S only

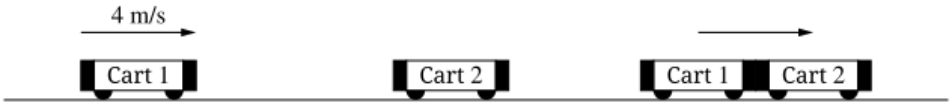
(D) At positions Q, R, and S

Answer A

Correct. Treating the collision as occurring instantaneously, the net external force due to the spring on the box results in zero impulse on the system, so the system's total momentum stays the same for all collision locations. However, since the collision is perfectly inelastic, there will be a change in kinetic energy unless the two objects are instantaneously at rest at the time of the collision. The objects are instantaneously at rest at position Q. However, at positions R and S, the spring has less potential energy than it does at Q, so there must be kinetic energy in the box-spring system. The box must be in motion, so the (nonzero) kinetic energy will change if the collision occurs at either R or S.

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30.



Two carts, Cart 1 of mass $0.25\ \text{kg}$ and Cart 2 of mass $0.50\ \text{kg}$, are free to move along a straight, horizontal track, as shown in the figure. There is negligible friction in the axles of the carts' wheels, and Cart 1 is initially moving to the right at $4\ \text{m/s}$ before colliding with Cart 2, which is initially at rest. After the collision, the carts stick together and move to the right. What is the kinetic energy of the two-cart system immediately after the collision?

(A) $\frac{2}{3}\ \text{J}$ ✓

(B) $1\ \text{J}$

(C) $2\ \text{J}$

(D) $6\ \text{J}$

Answer A

Correct. Because the system's momentum is the same before and after the collision, $m_1 v_{1i} + m_2 \cdot 0 = (m_1 + m_2) v_f$, so $v_f = \frac{m_1}{m_1 + m_2} v_{1i} = \frac{4}{3}\ \text{m/s}$. The final kinetic energy is $K = \frac{1}{2}\left(m_1 + m_2\right) v_f^2 = \frac{1}{2}\left(0.75\ \text{kg}\right)\left(\frac{4}{3}\ \text{m/s}\right)^2 = \frac{2}{3}\ \text{J}$.

31. Identical carts X and Y are placed on a straight, horizontal track, where they are free to move with negligible friction in the axles of their wheels. The carts undergo two separate collisions. In Collision 1, Cart X moves to the right with an initial kinetic energy K_i toward Cart Y, which is at rest, and the carts collide elastically. After the collision, Cart X comes to rest, and Cart Y moves to the right. In Collision 2, Cart X again moves to the right with the same initial kinetic energy K_i toward Cart Y, which is at rest, but the carts stick together during the collision and then move together to the right. Is the speed of Cart Y greater immediately after Collision 1 or immediately after Collision 2 ? What reasoning supports the claim?

(A) Immediately after Collision 1. During Collision 1, Cart Y gains an amount of kinetic energy equal to K_i, while in Collision 2, Cart Y gains an amount of kinetic energy less than K_i. ✓

(B) Immediately after Collision 1. During Collision 1, Cart Y gains an amount of kinetic energy greater than K_i, while in Collision 2, Cart Y gains an amount of kinetic energy equal to K_i.

(C) Immediately after Collision 2. In Collision 1, the two-cart system loses more kinetic energy than in Collision 2.

(D) Immediately after Collision 2. When the carts stick together, Cart X transfers more energy to Cart Y than when the carts bounce off of one another.

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Answer A

Correct. Because Collision 1 is elastic, the total kinetic energy of the two-cart system does not change during the collision. Therefore, all of the kinetic energy of Cart X is transferred to Cart Y. In Collision 2, the collision is inelastic, so the kinetic energy decreases during the collision. Both carts move together, so the remaining kinetic energy in the system after the collision is split between Cart X and Cart Y.

32. Two blocks, one of mass m and one of mass $2m$, are free to move along a horizontal surface with negligible friction. The blocks undergo collisions in two separate scenarios. In the first scenario, the block of mass m moves with an initial speed v_0 toward the block of mass $2m$, which is initially at rest, and the blocks stick together after they collide. In the second scenario, the block of mass $2m$ moves with the same initial speed v_0 toward the block of mass m, which is initially at rest, and the blocks again stick together after colliding. The total kinetic energy of the two-block system decreases by an amount ΔK_1 in the first scenario and by ΔK_2 in the second scenario. What is the ratio $\frac{\Delta K_2}{\Delta K_1}$?

(A) $\frac{1}{3}$

(B) $\frac{1}{2}$

(C) $\frac{2}{3}$

(D) 1 ✓

Answer D

Correct. The change in kinetic energy is the same in both scenarios. For the first scenario, conservation of momentum implies that $mv_0 = 3mv_1$, so that $v_1 = \frac{v_0}{3}$, and $\Delta K_1 = \frac{1}{2}mv_0^2 - \frac{1}{2}3m\left(\frac{v_0}{3}\right)^2 = \frac{1}{3}mv_0^2$. For the second scenario, conservation of momentum implies that $2mv_0 = 3mv_2$, $v_2 = \frac{2v_0}{3}$ and $\Delta K_2 = \frac{1}{2}2mv_0^2 - \frac{1}{2}3m\left(\frac{2v_0}{3}\right)^2 = \frac{1}{3}mv_0^2$.

33. Object 1 of mass m and Object 2 of mass $3m$ can slide along a straight, horizontal track with negligible friction. Object 1 initially moves to the right with speed v toward Object 2, which is initially at rest, and the two objects collide. Immediately after the collision, Object 1 moves to the left with speed $\frac{v}{2}$. Is this information sufficient to determine whether the collision was elastic? Why or why not?

(A) Yes, because the fact that the velocity of Object 1 changed direction during the collision means that the collision was not elastic.

(B) Yes, because conservation of momentum can be used to determine the final speed of Object 2, and the total kinetic energy of the two-object system after the collision can be calculated. ✓

(C) No, because the final speed of Object 2 cannot be determined without knowing whether the collision was elastic or inelastic.

(D) No, because the collision can be elastic for many different values of the final speed of Object 2.

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Answer B

Correct. The total momentum of the two-object system is constant during the collision. Therefore, conservation of momentum can be used to find the speed v_2 of Object 2 after the collision:
$\overrightarrow{p}_i = \overrightarrow{p}_f$
$mv = m\left(-\frac{v}{2}\right) + (3m)v_2$. To determine whether the collision was elastic, the initial total kinetic
$v_2 = \frac{v}{2}$
energy can be compared to the final total kinetic energy. If they are equal, the collision was elastic.
$K_i = \frac{1}{2}mv^2$
$K_f = \frac{1}{2}m\left(-\frac{v}{2}\right)^2 + \frac{1}{2}(3m)\left(\frac{v}{2}\right)^2 = \frac{1}{2}mv^2$.
$K_i = K_f$

34. An 8-kg object is moving with a speed of $6\ \text{m/s}$ along a horizontal surface while a 4-kg object is moving in the opposite direction with a speed of $3\ \text{m/s}$ on the same surface. The objects collide and stick together after the collision. Friction between the objects and the surface is negligible. What is the decrease in total kinetic energy of the two-object system during the collision?

(A) 54 J
(B) 90 J
(C) 108 J ✓
(D) 162 J

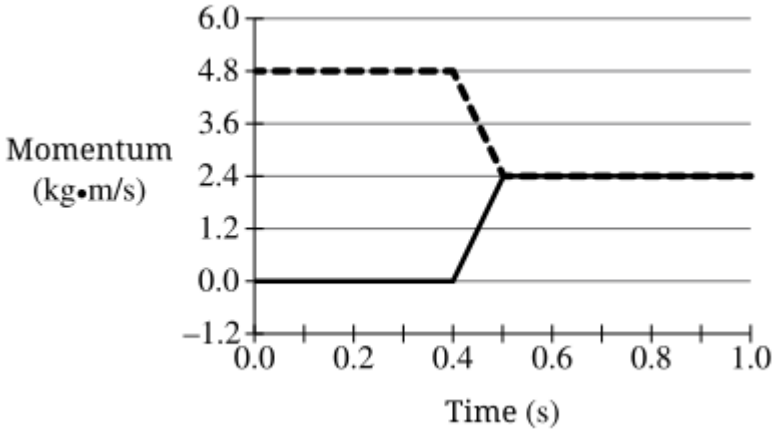
Answer C

Correct. Setting $m_1 = 8\ \text{kg}$, $v_1 = 6\ \text{m/s}$, $m_2 = 4\ \text{kg}$, and $v_2 = 3\ \text{m/s}$, conservation of momentum, $m_1v_1 + m_2v_2 = (m_1 + m_2)v$, determines the speed of the objects when stuck together, $v = 3\ \text{m/s}$. The kinetic energy lost in the collision is then $\frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 - \frac{1}{2}\left(m_1 + m_2\right)v^2 = 108\ \text{J}$.

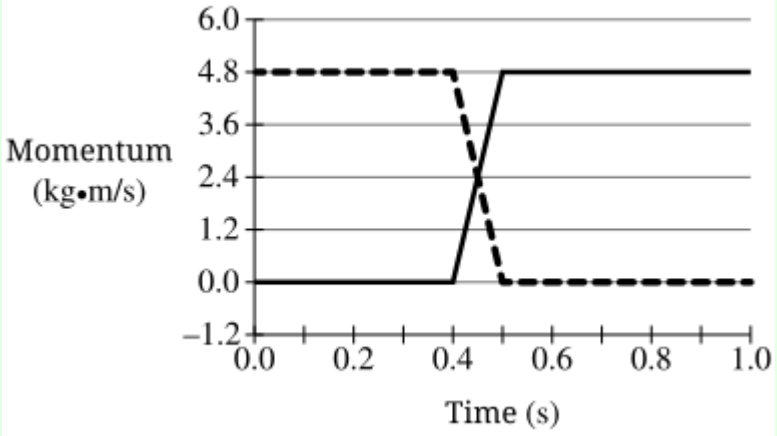
35. Two objects of equal mass undergo a series of collisions. For each collision, a graph is created of the momenta of both objects as functions of time. Which of the following graphs represents an elastic collision?

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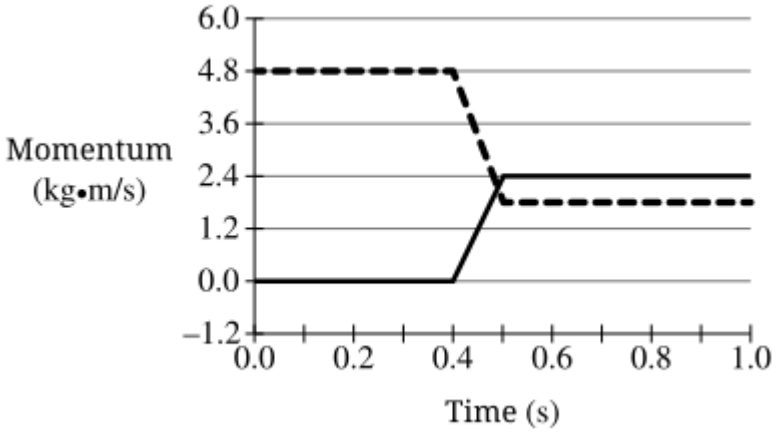
(A)



(B) ✓

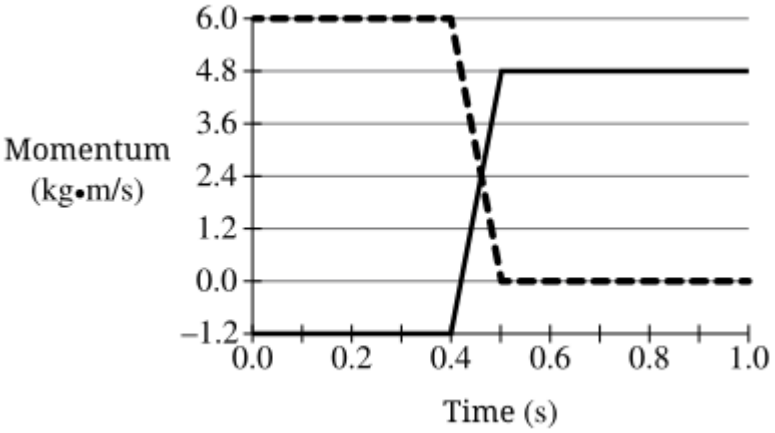


(C)



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(D)

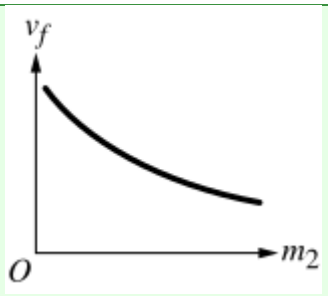


Answer B

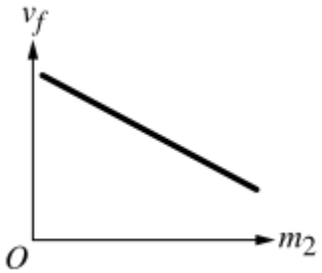
Correct. Both before and after the collision, one of the objects is at rest and one is moving. The object that is moving has the same momentum value and, since the objects also have the same mass, the moving object has the same velocity and therefore the same kinetic energy as well. Since one object is at rest, the total kinetic energy of the two-object system is the kinetic energy of whichever object is moving and therefore is the same before and after the collision. Since the system's kinetic energy is the same, the collision is by definition elastic.

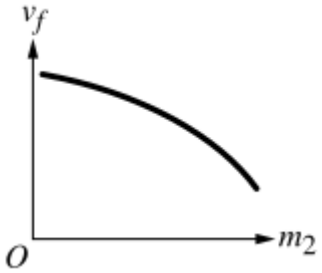
36. Students perform an experiment with multiple trials using blocks that can slide with negligible friction on a straight, horizontal track. In each trial, a block of mass m_1 slides with speed v_1 on the track and collides with a stationary block of mass m_2. The blocks stick together and move with speed v_f after the collision. In each trial, m_1 and v_1 are kept constant but the mass m_2 of the stationary block is varied, and v_f is recorded. The students graph v_f as a function of m_2. Which of the following could represent a best-fit curve to the data collected by the students?

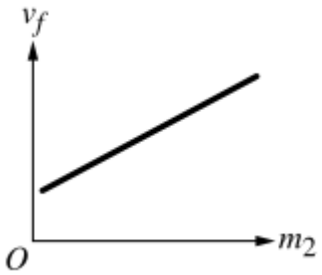
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(A) 

✓

(B) 

(C) 

(D) 

Answer A

Correct. Using conservation of momentum, the relationship between v_f and m_2 can be determined
$p_i = p_f$
$m_1 v_1 = (m_1 + m_2) v_f.$
$v_f = \frac{m_1}{m_1+m_2} v_1$
Therefore, v_f is inversely proportional to the sum of m_1 and m_2. Because m_1 is constant, as m_2 increases the value of v_f will decrease and asymptotically approach zero.